

Homework 1

- Q1.** What is the variable in the expression $2x + 3$?
(A) 2
(B) 3
(C) x
(D) $2x$
(E) $2 + 3$
- Q2.** Identify the coefficient in the term $4y$
(A) y
(B) 4
(C) $4y$
(D) 1
(E) 0
- Q3.** What is the constant term in the expression $2x - 5$?
(A) $2x$
(B) -5
(C) 2
(D) 5
(E) $-2x$
- Q4.** In the expression $3 + 2$, what is the operator?
(A) 3
(B) 2
(C) +
(D) -
(E) *
- Q5.** What does the term $2x^2$ represent?
(A) The coefficient of x is 2 and the exponent is 2
(B) The variable is 2 and the coefficient is x^2
(C) The coefficient of x is 2 and x is squared
(D) The coefficient is x and the variable is 2
(E) The constant term is 2 and the variable term is x^2
- Q6.** In the expression $x + 2$, what operation is being performed?
(A) Subtraction
(B) Addition
(C) Multiplication
(D) Division
(E) Exponentiation
- Q7.** Identify the term that is a variable in $3x + 2y - 4$
(A) $3x$ and $2y$
(B) $2y$ only
(C) $3x$ only
(D) -4
(E) $3x + 2y$
- Q8.** What is the purpose of the order of operations?
(A) To simplify fractions
(B) To solve equations
(C) To evaluate expressions with multiple operations
(D) To graph functions
(E) To find the perimeter of a shape

Open Ended Questions (FRQ)

- Q9.** Draw and explain how you would represent the expression $2x + 1$ using algebra tiles. Be sure to include a key for your drawing.
- Q10.** Explain, in your own words, what a variable is in an algebraic expression. Provide an example of a simple expression that contains a variable and describe what the variable represents in the context of the expression.

Chef's Tips

- Focus on introducing basic concepts and definitions related to variables and expressions, such as coefficients and constants.
- Use simple, one-step problems to help students understand the fundamentals of the order of operations.
- Encourage students to use visual aids like algebra tiles to represent expressions and enhance their understanding of variables and expressions.